

Muhammad Hamza

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Github: <https://github.com/m-hamza7> **Portfolio:** <https://muhmd-hamza-portfolio.vercel.app/>

PROFESSIONAL SUMMARY

Detail oriented and motivated Computer Science student with a strong foundation in Software Quality Assurance principles and the Software Testing Life Cycle (STLC). Hands-on experience in manual testing using JIRA (AIO Tests) and basic test automation using Playwright, including script development for web application testing. Familiar with test case design, bug reporting, and quality standards. Demonstrates strong attention to detail, analytical thinking, and a commitment to delivering reliable software solutions. Eager to begin a career in SQA and contribute to high-quality product development.

EDUCATION

FAST NUCES | 2022-2026

Bachelor of Science in Computer Science

SKILLS

Programming Languages

- Python
- C
- C++ (OOP)
- JavaScript
- Assembly (8086)

Web Development

- HTML5 / CSS3
- JavaScript (ES6+)
- React.js
- Flask (Python)
- Responsive UI Development
- Landing Page

Backend & Databases

- RESTful APIs
- MySQL / SQL
- CRUD Development
- Authentication & Authorization
- Server-Side Routing (Flask)

SQA

- Manual Testing
- Automation Testing (basics)
- STLC Understanding
- API Testing (basics)

Machine Learning / AI (BASICS)

- Model Training Dataset Labeling &
- Preprocessing Checkpoint
- Management & Model Evaluation

Tools & Platforms

- Jira (AIO Tests)
- Playwright
- Postman
- MS Dynamics 365
- MS Power Platform
- VS Code
- Git/Github
- MS Office

Software Engineering

- SDLC & STLC Understanding
- UML Modeling
- Version Control (Git/GitHub)
- Debugging & Optimization
- File Handling (C/C++)

PROFESSIONAL EXPERIENCE

Freelance Projects

PPC Landing Page Developer

- Designed and developed 100+ high-converting PPC landing pages and mini-websites for clients, focusing on fast performance, enhanced UI/UX, clear CTAs, and optimized user flows to maximize ad ROI.

ACADEMIC PROJECTS

CleanAI- Final Year Project (Ongoing) Website: <https://cleanai-project.vercel.app/>

AI-Powered Waste Classification System — Python, YOLO, Flask

Developing an intelligent waste-sorting system using deep-learning-based object detection (YOLO). Built a prototype demonstrating the Mechanism of the Final App and designed a polished landing page showcasing the project overview, features, and demo flow. Currently training and refining the detection model to improve accuracy and real-world performance, and designing Admin dashboard.

Online Blood Donation Management System — Python (Flask), MySQL

Repo Link: <https://github.com/m-hamza7/Online-Blood-Donation-System>

- Developed a complete web-based MIS for managing donors, hospitals, appointments, and blood inventories. Implemented role-based access, CRUD operations, secure authentication, and clean UI templates.

Atomic Chess (Chess variant) — React, JavaScript (AI-Based Game Logic)

<https://github.com/m-hamza7/Atomic-Chess>

- Developed an interactive web-based version of Atomic Chess, implementing custom game rules, move validation, and explosion mechanics. Built the UI in React with clean state management and smooth piece animations. Integrated basic AI logic for automated opponent moves and tested gameplay for stability and rule accuracy.

Library Management System — C++ (OOP and DS)

- Designed a modular library system featuring book inventory, member records, and borrowing/return workflows, applying object-oriented principles and file handling.

Poker Game — C

- Developed a terminal-based poker game featuring card shuffling, betting logic, and winner determination using data structures.

Regex-Based Search Engine — Python

- Implemented a lightweight text search tool using regular expressions to match patterns across documents with efficient parsing and filtering.

Hangman Game — Assembly Language (8086)

https://github.com/m-hamza7/HANGMAN-IN-ASSEMBLY_LANGUAGE

- Built a fully interactive Hangman game using low-level assembly instructions, handling user input, string manipulation, and game-state logic.
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